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RAIPAL: A Powerful and Agile Humanoid Arm for Human Tasks

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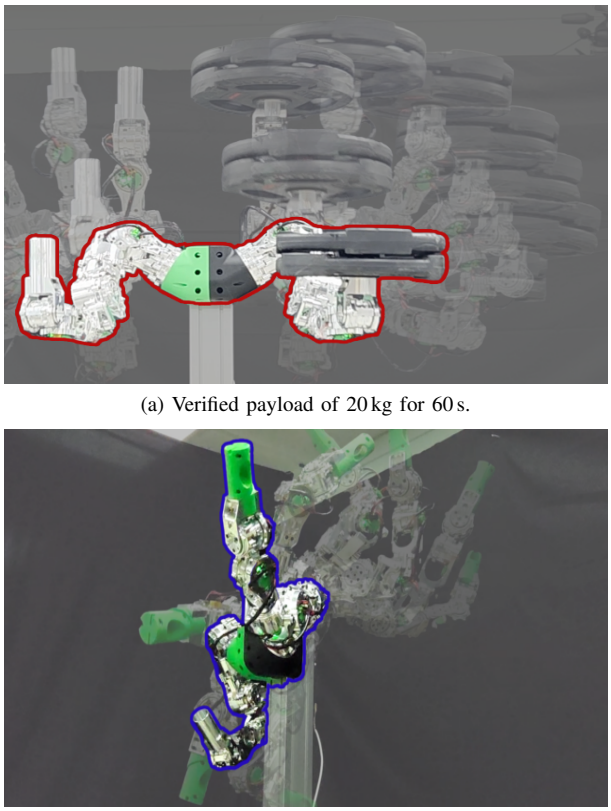
***Abstract*— This paper presents humanoid manipulator hardware designed specifically for human-level performance throughout its broad domain of tasks, from heavy payloads to highly dynamic motions. Unlike conventional designs targeting uniform workspace capability, the proposed approach derives force, speed, and endurance targets from published measurements of task-relevant human upper-limb performance. These targets guide the actuator, mechanism, and thermal design toward torque transparency and sustained operation. The platform integrates force-oriented upper-arm components, speed-oriented forearm components, and a posture-dependent elbow mechanism that shifts its torque-speed behavior. These components are driven by planetary actuators and compact actuator-embedded electronics. Experimental results validate broad task capability, demonstrating a peak payload of 20 kg (2.25× its own mass), an endpoint speed of 20.6 m/s, and continued operation under repeated hammering impacts. Thermal analysis further confirms that the robot's sustained operation capability exceeds the published human load–duration boundary across nearly the entire load spectrum. These findings show that hardware targets derived from human-task measurements narrow the human-to-humanoid capability gap. A supplementary video is available at https://youtu.be/xVUWFICD5G0**

***Index Terms*—Humanoid Robot Systems, Product Design, Development and Prototyping, Actuation and Joint Mechanisms.**

I Introduction

ADVANCES in permanent-magnet motors and specialized robotic actuators let modern humanoids surpass average human benchmarks in isolated capabilities [1], [2]. However, payload and fast motion typically reflect different hardware priorities on different platforms, as shown in Table I. General-purpose humanoids emphasize payload and runtime, whereas speed-oriented systems remain specialized and impractical for basic payload requirements. Characteristics of human upper-limb tasks instead reveal a coordinated capability envelope spanning heavy-duty manipulation, rapid distal motion, compliant interaction, and load-dependent endurance [3]. The direct use of these task priors and quantitative boundaries as hardware-design targets, however, remains largely unexplored.

Conventional manipulator sizing projects worst-case task-space forces into joint-space torque requirements, enforcing payload capability across broad workspace regions [4]. These guarantees suit fixed-base industrial manipulators but poorly match human tasks realized through coordinated, posture-dependent motion. At the actuator level, the same bias is reinforced by conventional transmission choices: high-reduction precision drives favor compact torque generation but penalize speed, reflected inertia, and torque transparency, whereas low-reduction quasi-direct-drive actuators favor backdrivability and dynamic interaction but require larger motors for heavy-duty tasks proportional to human arms [5]. As a result, conventional sizing and actuator tradeoffs tend to push humanoid arms toward either *strong and slow* or *fast and weak* designs, rather than broad human-task capability.



(a) Verified payload of 20 kg for 60 s.

(b) Verified endpoint speed of 20.6 m/s.

Fig. 1. Representative demonstrations. High-payload manipulation and high-speed upper-limb motion achieved with the same humanoid arm.

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TABLE I
COMPARISON OF KEY HARDWARE AND TASK-LEVEL CAPABILITIES ACROSS SELECTED STATE-OF-THE-ART PLATFORMS

Metric	Proposed Arm	KUKA-DLR LWR [6]	D3-ARM [7]	IIT Bi-Manual [1]	ARMADA [5]	LIMS [8]	AthenaZero [2]	DynaArm [9]
DoF (one arm)	7	7	6	7	6	7	7	6
Mass (one arm)	8.9 kg	15.0 kg	16.5 kg	9.0 kg	1.09 kg*	5.84 kg	N/A	9.2 kg
Peak Payload	20.0 kg	N/A	2.0 kg	15.0 kg	2.5 kg	3.0 kg	3.0 kg	12.0 kg
Continuous Payload	5.0 kg	7.0 kg	N/A	7.0 kg	N/A	N/A	N/A	6.0 kg
Peak Payload-to-Arm-Mass Ratio	2.25:1	N/A	0.12:1	1.67:1	2.29:1*	0.51:1	N/A	1.30:1
Max Speed	20.6 m/s	N/A	1.47 m/s	N/A	6.16 m/s	5.35 m/s	31.0 m/s†	10.0 m/s

* Reported mass only includes the moving parts, without body-mounted components and the gripper
† Maximum throwing velocity achieved through coordinated torso-waist dynamic trajectory optimization [2]
Scope: This comparison includes peer-reviewed systems and commercial platforms with public video demonstrations
N/A indicates that the corresponding specification was not reported in the cited source

This paper proposes a 7-DoF humanoid manipulator built for the wide domain of practical human tasks. Instead of enforcing uniform workspace-wide capability, the proposed approach extracts engineering targets from published measurements and analyses of task-relevant human upper-limb force, speed, posture, and endurance. The resulting prototype, shown in Fig. 1, is validated through real-world experiments and analysis. The key contributions of this work are as follows:

- A source-grounded method for deriving robot force, speed, endurance, and mass-distribution requirements from measurements of human upper-limb tasks.
- A top-down methodology that maps those task-derived parameters into force-, speed-, and posture-specialized hardware components.
- Experimental validation through high-payload, high-speed, endurance, and high-impact demonstrations.

II METHODS

II-A The Human Prior

Published evidence and measurements of human upper-limb tasks provide the task-relevant engineering parameters used to design this robot platform. These task characteristics span high-force and high-speed motion, mass distribution, posture-dependent behavior, compliant interaction, and load-dependent endurance, which together define the intended domain of a high-performing humanoid arm. Rather than reproducing human anatomy, these task characteristics are translated into robot design requirements such as endpoint force, joint speed, torque transparency, and sustained operation.

Feedback-Driven Accuracy: Characteristics of contact-rich human tasks indicate that safe and accurate interaction combines mechanical compliance with continuous feedback rather than relying on extreme stiffness [3]. This compliance enables the arm to yield under contact while maintaining precision through closed-loop correction. Consequently, the mechanical design should prioritize torque transparency and low drivetrain impedance so that feedback-based interaction control is not hindered by excessive mechanical resistance.

Upper-Arm Strength: Characteristics of high-force upper-body tasks contract the arm posture to shorten the external moment arm about the proximal joints [10]. This posture uses joint constraints to support large forces through the upper-arm. The forearm remains nearly parallel to the force, transmitting proximal force while requiring mainly stabilizing torque.

Forearm Acceleration: Characteristics of high-velocity upper-limb tasks instead include an extended arm posture that increases the radius of rotation. The forearm accelerates its low-inertia distal segments to generate wrist-snap motion [11].

Elbow Mediation: The elbow mediates between the upper-arm and forearm, and must support both high-torque and high-velocity tasks. The contracted high-torque and extended high-speed patterns require the elbow to provide high torque in flexion and high velocity in extension. In the human elbow, the interplay of moment arms and muscle coordination realizes this behavior, producing larger torque when flexed and higher velocity when extended [12].

Endurance: Published sustained-exertion measurements show that human upper-limb task endurance varies strongly with applied load. For humanoid arms, this task-derived load–duration boundary translates into a thermal requirement because heat accumulation in the actuators and embedded electronics limits the duration of sustained joint torque. Rohmert's maximum-endurance-time relation is relevant because it provides a normalized human-task endurance boundary that can be compared directly with the actuator thermal Safe Operating Area (SOA) [13]. The robot thermal SOA should therefore support sustained operation over the relevant load spectrum.

II-B Human-Task-Derived Engineering Parameters

The numerical targets in Table III are representative rounded adult-male design values rather than physiological limits. Each task-specific target uses the reported cohort mean plus one standard deviation ($\mu + \sigma$), corresponding to approximately the upper 16% under a normal-distribution approximation; this defines a concrete above-average benchmark. Reported measurements are mapped and rounded to the corresponding robot axes used in Table III. The maximum-torque inputs are taken directly from axis-specific shoulder, elbow, wrist flexion–extension and deviation, and forearm-rotation measurements [14]–[16]. The peak-speed inputs are mapped and rounded from reported maximum upper-limb joint velocities in sports actions [17]. The 1.80 m, 80 kg human model provides the common basis for converting reported segment proportions to link lengths and masses and body-mass-normalized torque measurements to peak joint torques, with the resulting model supplying the gravity terms [3].

Endpoint Force: Conventional worst-case task-space forces can be projected to joint-space through Jacobian-transpose and gravity terms as follows [4]:

$$\tau_{\max,i} \geq \beta \max_{q} \left\| J(q)_i^T f_{\text{ext}} + g(q)_i \right\| \tag{1}$$

where q is the generalized pose, $J(q)$ is the Jacobian, f_{ext} is the external force objective, $g(q)$ is the generalized gravity term, and β is a safety factor. Applying this projection as a uniform workspace constraint can narrow the humanoid-arm task domain, but the projection remains useful for converting task-space objectives into joint-space requirements. Reversing this projection gives endpoint force, a more relevant heavy-duty metric than direct torque matching, which would oversize proximal links and the elbow actuator. For each joint, endpoint force is defined as the minimum gravity-compensated end-effector force achievable across configurations:

$$f^H_{\text{end},i} := \min_{q \in \mathcal{Q}_H} \left\| J^H_i(q)^{-T} \left(\tau^H_{i,\max} - g^H_i(q) \right) \right\| \tag{2}$$

where $\mathcal{Q}_H$ is the evaluated human configuration set, and $J^H_i(q)$ and $g^H_i(q)$ are the Jacobian contribution and gravity term of the anthropometric model.

A worked example is provided for Axis 1. Lategan reports a body-mass-normalized shoulder-extension torque of 1.23 ± 0.21 Nm/kg [14]. Applying $\mu + \sigma$ to the 80 kg reference gives a maximum joint torque of $\tau^H_1 = (1.23 + 0.21)(80) = 115.2$ Nm. Using upper-arm, forearm, and hand length fractions of $0.186H$, $0.146H$, and $0.108H$, and mass fractions of $0.028M$, $0.016M$, and $0.006M$, the 1.80 m, 80 kg model has a 0.7920 m total reach, a 4.00 kg total arm mass, and a 12.804 Nm worst-case gravity torque at full extension [3], [18]. This yields the endpoint-force projection

$$f^H_{\text{end},1} = \frac{115.2 - 12.804}{0.7920} = 129.3 \text{ N} \rightarrow 129 \text{ N}.$$

Similar projections are evaluated for each human motion and its corresponding worst-case pose.

Peak Velocity: Agile upper-limb capability is quantified by the maximum achievable angular velocity of each joint. The source values are measured peak joint velocities from compatible young-adult male cohorts, and the same $\mu + \sigma$ selection rule is applied. This joint-space metric is appropriate because unlike force requirements, velocity is kinematically determined by $v = J(q)\dot{q}$, allowing joint-speed targets to transfer directly to task-space speed given comparable topology and link proportions [4]. The resulting targets are later compared with real robot parameters in Table III.

Link mass and worst-case angular acceleration are retained as secondary constraints but are not part of Table III's primary endpoint-force and peak-speed comparison. Robot link masses are kept near 60% of the corresponding human segment masses, while upper-arm worst-case acceleration is evaluated separately as a secondary design check.

TABLE II
SPECIFICATIONS OF THE PROPOSED ACTUATOR FAMILIES

Feature	Upper-arm Actuator	Forearm Actuator
Mass	1.50 kg	0.36 kg
Bounding dimensions	$\phi 100 \times 83$ mm	$\phi 77 \times 60$ mm
Reduction	22:1 (3K compound)	7.33:1 (planetary)
Rated torque	56 Nm (30 min)	3.88 Nm (45 min)
Peak torque	180 Nm (26 s)	12.61 Nm (15 s)
No-load speed (80 V)	12.8 rad/s	89.7 rad/s
Reflected inertia	5.76×10^{-2} kg·m^2	5.41×10^{-4} kg·m^2

II-C Actuator Design

The four upper-arm joints use a 3K (also known as a Wolfrom gearbox) compound planetary transmission consisting of two axially aligned ring gears, rigidly connected compound planet gears, and a sun gear (shown in Fig. 2(a)) to provide torque amplification for high-payload manipulation while preserving practical backdrivability. The selected 22:1 reduction sits between high-reduction precision drives and low-reduction quasi-direct-drive actuators, avoiding the speed and reflected-inertia penalties of very high-reduction transmissions. The resulting actuator design, shown in Fig. 2(a), supports the force-oriented upper-arm notion in Table III.

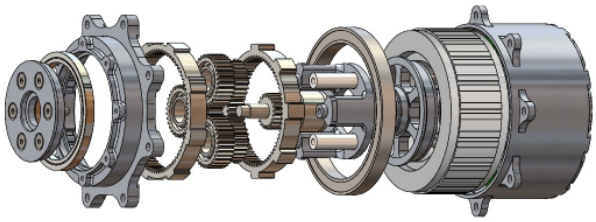
Gear preload trades backlash against static friction: tighter meshing reduces backlash but increases preload-induced friction. The final preload was selected empirically instead of through iterative tolerance adjustment. Section III-A evaluates the resulting backlash–friction tradeoff.

Gear-mesh Coulomb friction reduces transmission efficiency and torque transparency. The selected 3K compound planetary transmission is therefore modeled analytically and evaluated experimentally rather than optimized through tooth profiles that could compromise backlash or packaging [19]. Although serial two-stage planetary transmissions can be more efficient, the 3K topology integrates the required reduction into a compact humanoid upper-body package. A simpler stepped compound planetary could omit the second ring gear, but achieving 22:1 through the planet-stage size difference would enlarge the planet gears and overall gearbox envelope. To quantify transmission performance, forward- and backward-driving efficiencies are defined as

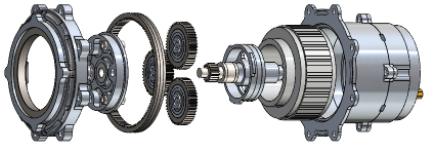
$$\eta_{\text{fwd}} = \frac{|\tau_{\text{out}}\,\omega_{\text{out}}|}{|\tau_{\text{in}}\,\omega_{\text{in}}|}, \quad \eta_{\text{bwd}} = \frac{|\tau_{\text{in}}\,\omega_{\text{in}}|}{|\tau_{\text{out}}\,\omega_{\text{out}}|} \tag{3}$$

where $(\tau_{\text{in}}, \omega_{\text{in}})$ and $(\tau_{\text{out}}, \omega_{\text{out}})$ denote the input- and output-side torque and angular velocity, respectively.

The forearm actuator in Fig. 2(b) prioritizes low reflected inertia, high speed, and backdrivability while meeting the distal torque targets from Section II-B. It uses a 7.33:1 single-stage planetary gearbox, which provides high joint speed and torque transparency for distal acceleration. Because single-



(a) Upper-arm actuator.



(b) Forearm actuator.

Fig. 2. Exploded actuator families with a 3K compound planetary upper-arm drive and a high-speed, low-inertia planetary forearm drive.

stage planetary transmissions are standard in proprioceptive robotic actuators [20], dedicated transmission analysis focuses on the less conventional 3K compound upper-arm drive.

II-D Mechanism Design

The mechanism is decomposed into a proximal force-specialized assembly, a distal speed-specialized assembly, and a posture-dependent elbow transmission. This follows from the task-dependent force–speed mediation in Section II-A. The platform terminates at the wrist mounting interface; articulated-hand design and task-specific parameterization are outside this work's scope.

The upper-arm is designed to prioritize torque while remaining as close as possible to the extracted upper-arm velocity targets. Four repeated high-torque modules add force capacity without joint-specific hardware complexity.

The forearm uses a serial arrangement of actuators to minimize distal inertia and prioritize speed while preserving the endpoint force required to stabilize heavy tasks. Although this arrangement may reduce accessibility during constrained-space manipulation compared with dedicated wrist mechanisms [8], it favors simplicity and rapid acceleration. The distal assembly therefore acts as the limb accelerator during snapping and throwing motions. At the implemented operating point, it exceeds all extracted wrist-speed targets while maintaining the required distal force margin.

The elbow implements the dual role described in Section II-A using a crossed-four-bar mechanism whose effective reduction varies with angle

$$r_{\mathrm{eff}}(\theta) = \omega_{\mathrm{in}}(\theta) \,/\, \omega_{\mathrm{out}}(\theta), \tag{4}$$

where θ is the elbow angle, and ω_{in} and ω_{out} are the input and output angular velocities. Similar posture-dependent joint mechanisms have previously been developed to tailor mechanical advantage to configuration-dependent functional requirements [21], [22]. As shown in Fig. 3, the implemented mechanism varies r_{eff} from 0.531 near extension to 0.875 near flexion, expanding the output speed region in extended postures and the output torque region in flexed postures. For simulation, a polynomial surrogate of the explicit formulation introduces negligible computational overhead. The linkage also shifts actuator mass proximally, reducing distal inertia and the effective inertia seen at the shoulder joints by 15%. The utility of this posture-dependent operating-region transformation is evaluated experimentally in Section III-C.

Table III compares the resulting robot arm design to the human-task-derived engineering targets set in Section II-B. The single-arm assembly weighs 8.9 kg, below an estimated 12.3 kg functionally matched human reference that includes 9.0 kg of base musculature driving similar motion but excludes the hand and extrinsic digit-actuating forearm musculature [23]. Despite the reduced mass, the system exceeds the proximal endpoint-force targets and achieves joint maximum speeds beyond the human reference. The realized shoulder axes provide 248–502 N of endpoint force, whereas the forearm axes provide 78.6–160.7 N while operating at substantially higher speeds, confirming the deliberate assignment of different force and speed objectives to the two arm regions.

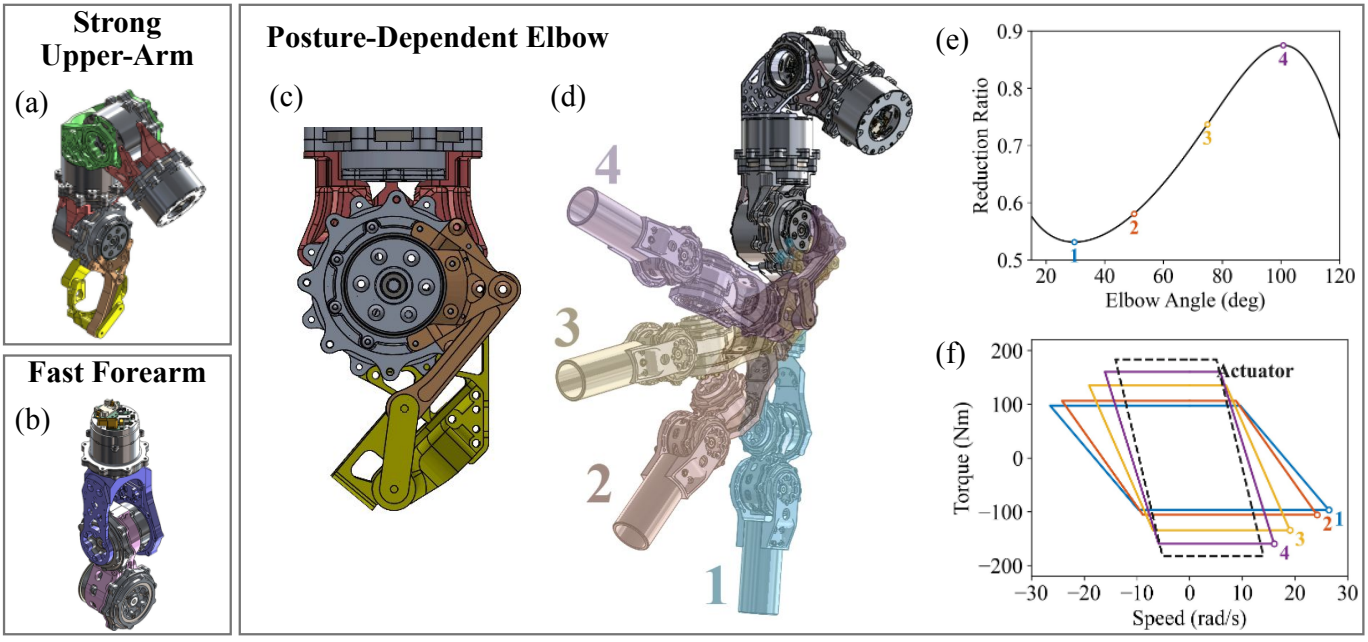


Fig. 3. Mechanism-level decomposition of the robot arm design. (a) The force-specialized upper-arm and (b) speed-specialized forearm implement the primary segment roles. (c,d) The posture-dependent crossed-four-bar elbow changes effective reduction with joint angle. (e,f) The resulting transformation expands torque capability in flexed postures and speed capability in extended postures.

TABLE III
COMPARISON AGAINST HUMAN-TASK-DERIVED ENGINEERING TARGETS

Joint Axis	Joint motion	Endpoint force (N)		Max speed (rad/s)	
		Target	Robot	Target	Robot[a]
1	Shoulder flex./ext.	129	287	9.0	12.8
2	Shoulder abd./add.	153	248	8.0	12.8
3	Shoulder rot.	145	502	12.0	12.8
4	Elbow flex./ext.	171	255	22.0	24.1
5	Wrist pron./sup.	84	89.3	23.0	89.7
6	Wrist flex./ext.	79	78.6	40.0	89.7
7	Wrist dev.	58	160.7	20.0	89.7

[a] Calculated at 80 V.

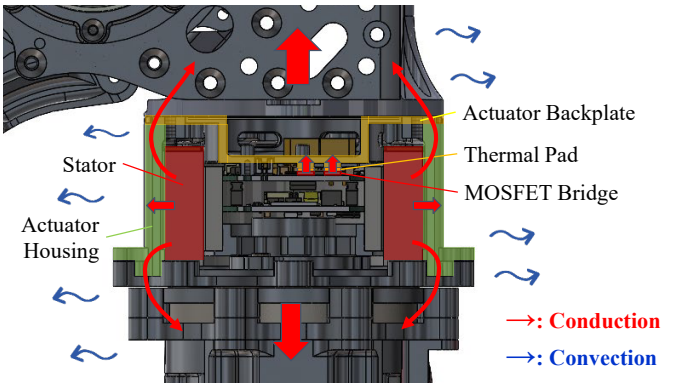


Fig. 4. Chassis conductive cooling paths of the integrated actuator. Red arrows indicate conductive heat flow from the stator and MOSFET bridge into the actuator chassis and attached robot-arm mechanism; blue arrows indicate convection from exposed surfaces.

II-E Electronics Design

Embedding a high-current servo drive inside each actuator places switching electronics close to sensing and communication circuits, making low-voltage signals vulnerable to noise. A stacked dual-board architecture separates the MOSFET power stage from the control-and-sensing board, preserving compact packaging while improving signal isolation across both actuator families. The layout thermally isolates control electronics from the MOSFET bridge.

Heat from the stator and MOSFET bridge is passively conducted into the actuator chassis through thermal pads and the motor housing, as shown in Fig. 4. The actuator chassis and attached robot-arm mechanism therefore form a shared passive heat path for continuous-current operation without dedicated airflow. Section III-B identifies and validates this thermal path.

III RESULTS AND EXPERIMENTS

III-A Upper-Arm Actuator Analysis

This experimental analysis verifies that the 3K compound planetary transmission satisfies the torque-transparency and packaging requirements in Sections II-B and II-C.

1) Backlash-Friction Tradeoff: A momentum-wheel experiment characterizes the relationship between transmission backlash and static friction. Low-amplitude torque perturbations reveal input-side backlash while the output is held quasi-static by a large-inertia wheel; the torque required to initiate motion at the same position provides the friction estimate. Measurements are repeated throughout loaded running-in.

Static friction decreases while input-side backlash increases during running-in, as shown in Fig. 5. This inverse relationship

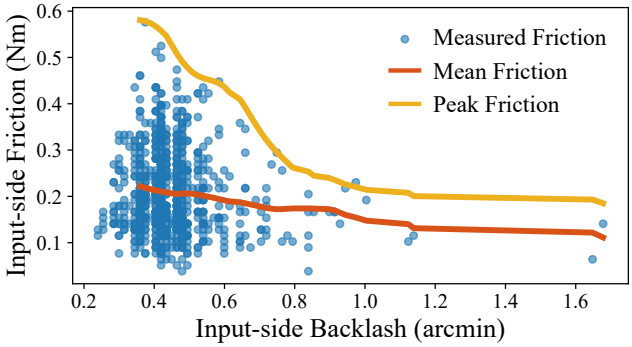


Fig. 5. Backlash-friction tradeoff of the upper-arm actuator. Minimum driving torque required to initiate motion is plotted against input-side backlash throughout a running-in process. Experimental samples and corresponding mean and peak friction trends show that static friction decreases as backlash increases, indicating strain-induced resistance in pre-loaded gear meshes.

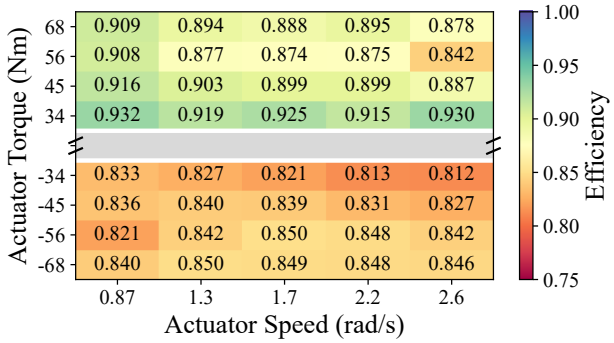


Fig. 6. Upper-arm actuator bilateral efficiency. The upper-arm actuator achieves an average efficiency of approximately 90% in forward driving and 84% in backward driving over the evaluated operating range.

suggests that a significant portion of static friction originates from elastic strain and contact deformation in preloaded gear meshes rather than simple Coulomb friction.

2) Bilateral Efficiency: A dynamometer couples two identical upper-arm actuators through a torque sensor, with one applying torque against the other's regulated load speed.

Fig. 6 shows the measured forward- and backward-driving efficiencies of the upper-arm actuator over the relevant operating range. Forward efficiency remains between 0.88 and 0.93, while the lower backward efficiency is consistent with directional efficiency asymmetry in geared transmissions [19].

III-B Endurance Bound Analysis

1) Thermal Identification: As described in Section II-A, sustained joint operation of humanoids is bounded by thermal limits. To identify the robot's thermal SOA, each joint was driven with constant current at 20 °C ambient temperature. The transient response was fit with a first-order model:

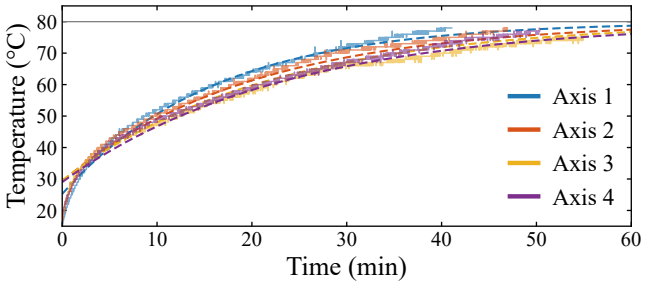
$$T(t) = T_{\text{amb}} + PR\left(1 - e^{-t/(RC)}\right), \tag{5}$$

where $T(t)$ is actuator temperature, T_{amb} is ambient temperature, P is total electrical loss, and R and C are the identified thermal resistance and capacitance [20]. The continuous power limit is computed as $P_{\text{allow}} = (T_{\max} - T_{\text{amb}})/R$, with continuous torque obtained from the actuator loss model. Table IV summarizes the identified parameters and continuous limits for each axis. Fig. 7 shows close agreement between measured and predicted temperatures, validating the model.

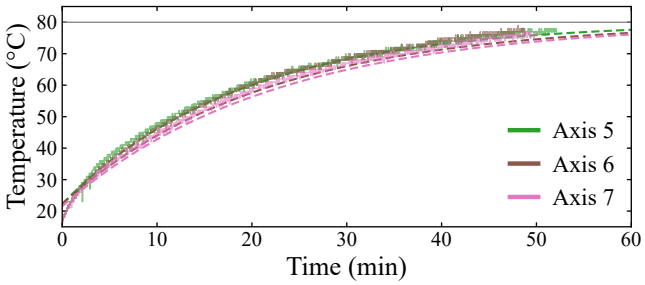
TABLE IV
AXIS-WISE THERMAL IDENTIFICATION AND CONTINUOUS OPERATING LIMITS OF THE INTEGRATED ROBOT ARM

Axis	Type	R (K/W)	C (J/K)	P_{allow} (W)	Cont. torque (Nm)
1	Upper-arm actuator	1.09	762.87	43.46	64.49
2	Upper-arm actuator	1.55	672.22	28.35	48.07
3	Upper-arm actuator	1.58	719.26	27.60	47.11
4	Upper-arm actuator	1.26	965.42	35.00	55.90
5	Forearm actuator	2.54	378.85	19.29	3.56
6	Forearm actuator	2.96	364.94	16.66	3.19
7	Forearm actuator	2.88	392.97	17.30	3.28

Note: P_{allow} assumes $T_{\max} = 80\ ^\circ\text{C}$; continuous torques are obtained from the actuator loss model. Axis-wise thermal parameters differ because attached mechanisms provide additional heat paths.



(a) Transient temperature response of upper-arm actuators (Axes 1–4).



(b) Transient temperature response of forearm actuators (Axes 5–7).

Fig. 7. Axis-wise transient temperature response. Solid curves show measured actuator temperatures; dotted curves show model predictions. Each joint was operated at its predicted continuous torque rating, and the measured temperatures were recorded over time.

2) Endurance Envelope: To evaluate the task-derived endurance target, the robot thermal SOA and the human endurance boundary are compared in a common joint-wise endurance–load space. Each joint load is mapped to a normalized human load coordinate as $\tau_i/\tau^H_{\max,i}$, where $\tau^H_{\max,i}$ denotes the corresponding source-derived, axis-mapped human torque reference defined in Section II-B. Rohmert's maximum-endurance-time relation maps normalized human static load to endurance-time, with loads below 15% treated as continuously sustainable [13]. Similarly, the robot thermal model predicts continuous operation below the normalized continuous-torque limit. Both thresholds appear as vertical asymptotes in Fig. 8.

At low and moderate loads, sustainable duration is governed by the RC model because heat can diffuse from the stator and power stage into the actuator chassis and arm structure. Conduction sets the robot's steady-state thermal boundary.

At high loads, Joule heating rises faster than conductive spreading to the chassis, so short-duration overloads are bounded by a quasi-adiabatic relation:

$$\left(\tau_i/\tau^R_{\max,i}\right)^2 t_{\text{sus}} = \text{const.}, \tag{6}$$

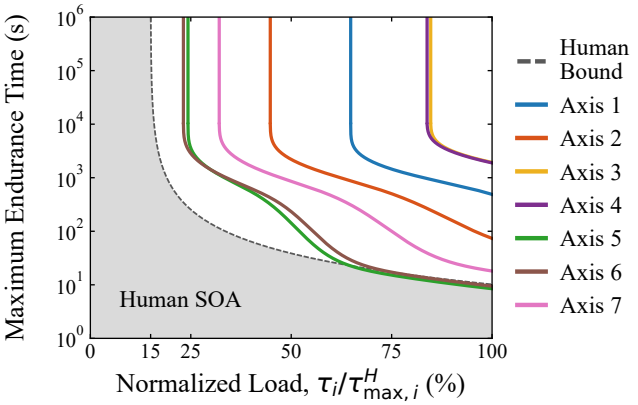


Fig. 8. Endurance envelope comparison in joint-wise endurance-load space. The gray region denotes the human SOA, and the regions below the colored boundaries denote the axis-wise robot thermal SOA.

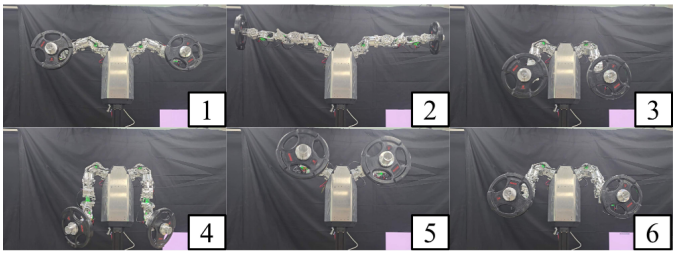


Fig. 9. Continuous payload demonstration sequence. The robot manipulates two 5 kg discs, one per arm, along a trajectory including worst-case poses for each joint, continuously for 25 minutes.

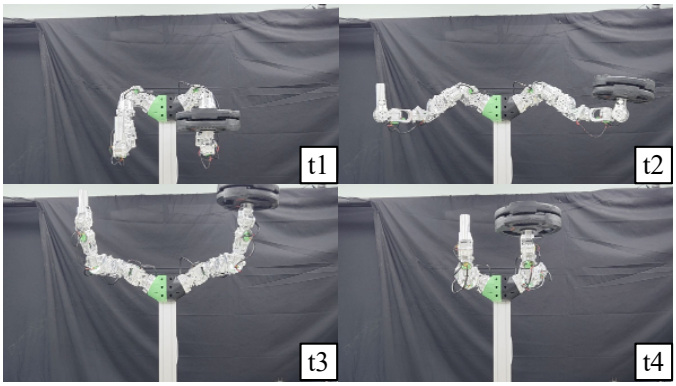
where $\tau^R_{\max,i}$ is the robot peak torque of axis i, and t_{sus} is the sustainable duration at the applied joint torque. For each axis, the final robot boundary in Fig. 8 is obtained by geometrically blending the conductive and quasi-adiabatic limits with a logistic weight constrained by empirical data.

As shown in Fig. 8, the robot thermal SOA largely encloses the human endurance envelope for all axes. Thus, the robot exceeds the human endurance envelope across the vast majority of the evaluated load range.

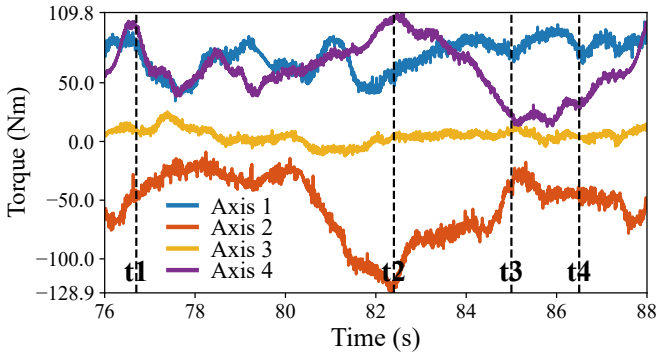
III-C Demonstrations

Four demonstrations evaluate whether the integrated arm can realize the broad task envelope targeted by the human-task-derived engineering parameters, spanning force, speed, and high-impact tasks. Each reported demonstration was repeated up to ten times, reproducing the reported qualitative outcome or quantitative result without maintenance between trials or observable damage to the hardware components. All experiments used manually designed joint-space trajectories tracked by a joint-level impedance controller running at 15 kHz, with trajectory targets updated at 2 kHz and fixed PD gains used across all demonstrations.

1) Continuous Payload: In the continuous-payload demonstration shown in Fig. 9, the robot manipulated 5 kg per arm for 25 minutes along a trajectory passing through the worst-case pose for each joint, while all actuator temperatures remained below the prescribed limit. This provides task-level evidence of repeated, thermally safe manipulation rather than a standalone endurance measure.



(a) Peak payload demonstration sequence.

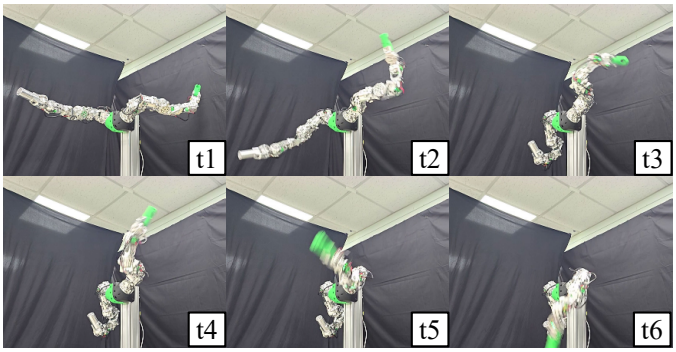


(b) Upper-arm torque data during peak payload demonstration.

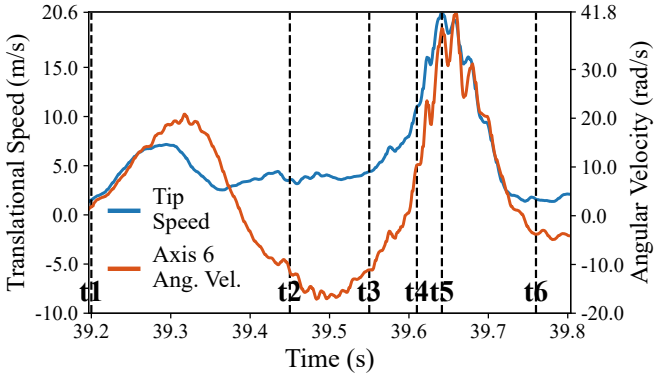
Fig. 10. Peak payload demonstration. The robot manipulates two 10 kg discs with one hand for 60 seconds. The trajectory resembles human lifting with neutral forearm positions, while also including an outward extension to probe a workspace wider than the typical lifting motion.

2) Peak Payload: To evaluate short-burst force capability, the robot repeatedly manipulated a 20 kg payload with one arm for 60 seconds as shown in Fig. 10(a). The trajectory included the extended posture $t2$, which is more demanding than the contracted postures typically used in human lifting and therefore provides a conservative verification of peak force capability. Fig. 10(b) confirms that the upper-arm actuators operate in their high-torque regions, particularly at $t2$, despite the lower-torque forearm axes. This experiment validates coordinated high-force operation across the robot task envelope. In particular, the Axis 4 elbow-torque trace exceeds $100\ \mathrm{Nm}$, showing that a fixed low reduction selected only to match the elbow speed used during pitching in Section III-C.3 would be insufficient for the demonstrated high-payload motion.

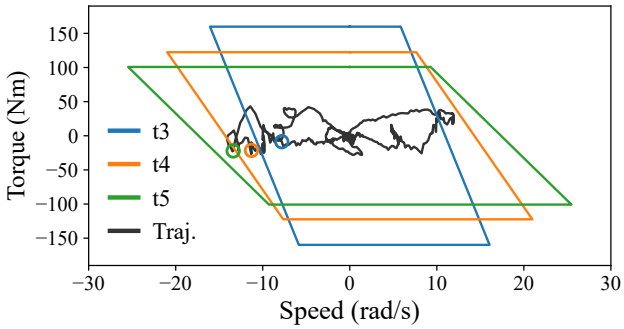
3) Pitching: The pitching demonstration reached a peak endpoint speed of $20.6\,\mathrm{m/s}$, approximately 130% of the average human reference. Fig. 11(b) confirms rapid distal acceleration near the peak-velocity point $t5$. Furthermore, torque-speed analysis in Fig. 11(c) shows that the elbow operating region expands speed-wise as the elbow extends into the peak-velocity posture $t5$. These measurements show that the endpoint-speed result arises from the intended directive for distal acceleration and posture-dependent speed expansion rather than an isolated actuator peak. This verifies coordinated athletic motion through posture-dependent elbow action.



(a) Pitching demonstration sequence.



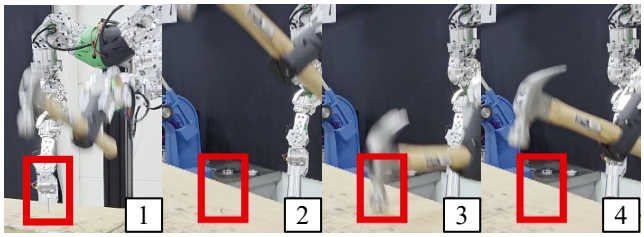
(b) Endpoint speed and axis 6 (forearm flexion) angular speed.



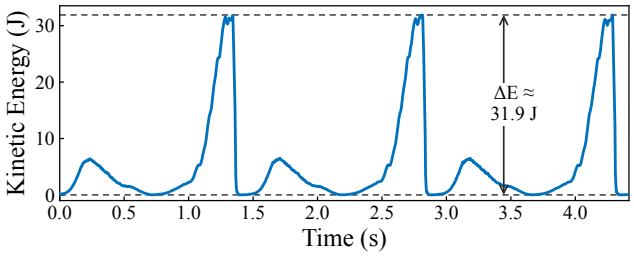
(c) Elbow operating regions (parallelograms) and pitching torque–speed trace (points) at representative times.

Fig. 11. Pitching demonstration. The fast forearm and posture-adjusted elbow operating region produce a peak endpoint speed of 20.6 m/s.

4) Hammering: The hammering demonstration shown in Fig. 12 evaluates whether the complete arm can sustain repeated impact-rich tool use and continue operating without maintenance or observable structural damage. The task uses constant control gains. Seven consecutive strikes per nail were repeated on ten nails (70 total impacts) without intervening maintenance or observable hardware damage. The hammer assembly had a mass of $0.753\ \mathrm{kg}$ and an overall length of $0.312\ \mathrm{m}$. Its average impact velocity over the 70 strikes was $3.8\ \mathrm{m/s}$, with an average estimated hammer contact impulse of $14.6\ \mathrm{N\,s}$. The robot-only complete-left-arm kinetic energy was logged throughout three representative strikes, with the hammer mass and inertia excluded; Fig. 12(b) shows approximately 32 J of robot-side kinetic energy dissipated nearly instantaneously at each impact. The repeated energy dissipation through impacts relevant to human tasks demonstrates human-level capability in contact-rich settings.



(a) Hammering demonstration sequence.



(b) Robot-only whole-left-arm (hammer excluded) kinetic energy over three instrumented strikes.

Fig. 12. Hammering demonstration sequence. The robot drives ten pre-positioned nails with seven strikes each (70 total), without maintenance or observable hardware damage. The red box highlights the tacked nail. Three instrumented strikes show approximately 31.9 J of robot-only arm kinetic energy (hammer excluded) dissipated at impact. ‘

IV Conclusion

This paper presented a humanoid manipulator built for human tasks by deriving engineering targets from published task characteristics and measurements rather than imposing uniform workspace-wide sizing. This process yielded a force-specialized upper-arm, a speed-specialized forearm, a posture-dependent elbow mechanism, heterogeneous proximal and distal actuators, and compact actuator-embedded electronics with chassis conductive cooling. Experiments verified the resulting hardware through a 20 kg peak payload, 20.6 m/s endpoint speed, endurance exceeding the human endurance envelope across nearly the entire load range, and high-impact demonstrations. Together, these results show that the developed arm combines force, speed, endurance, and physical interaction capabilities within a single integrated humanoid manipulator. The results suggest that humanoid-arm capability is shaped not only by actuator technology, but also by hardware requirement choices. By using human-task-derived engineering targets, the proposed approach distributes capability across specialized arm regions instead of oversizing every joint for uniform workspace guarantees. This provides a practical route toward humanoid manipulators that cover a broader human-task envelope without being forced into specialized regimes or devising complex energy-storing mechanisms.

Future Work: A natural next step is to integrate the arm into a full humanoid and explore challenging whole-body locomotion and manipulation tasks. Its low reflected inertia, broad task capability, and torque-transparent actuation also motivate more sophisticated control methods for a broader range of human tasks.

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